

Figure6C

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Intro

Temperature range analysis of taxa between the Summer–Autumn and Winter–Spring semesters, to reproduce the results of Figure 6B from the original publication Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
```

2. Load data

asvs_workable.tsv contains the rarefied ASV abundance profiles, with samples as rows and ASVs as columns.

samo_metadata_workable.tsv contains the metadata for the samples. date2season2community.tsv is a table mapping the date, season, and community columns.

```
ABUND <- read_tsv("../data/asvs_workable.tsv.gz", show_col_types = FALSE)
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUNITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Convert ABUND to long format and map metadata

```
ABUND_long <- ABUND %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(~Date, names_to = "asv_id", values_to = "abund") %>%
  left_join(x = .,
    y = METADATA %>% dplyr::select(Date, sample_name, Temperature_C),
    by = c("Date")) %>%
  left_join(x = .,
    y = DATE2SEASON2COMMUNITY ,
    by = "Date") %>%
  filter(abund > 0) # it is critical to remove zeros, other wise asv ranges in S1 and S2 are
```

4. Compute temperature range for each ASV

```
asvs2range <- ABUND_long %>%
  group_by(asv_id) %>%
  mutate(max_temp = max(Temperature_C),
    min_temp = min(Temperature_C)) %>%
  mutate(asvs_temp_range = max_temp - min_temp)
```

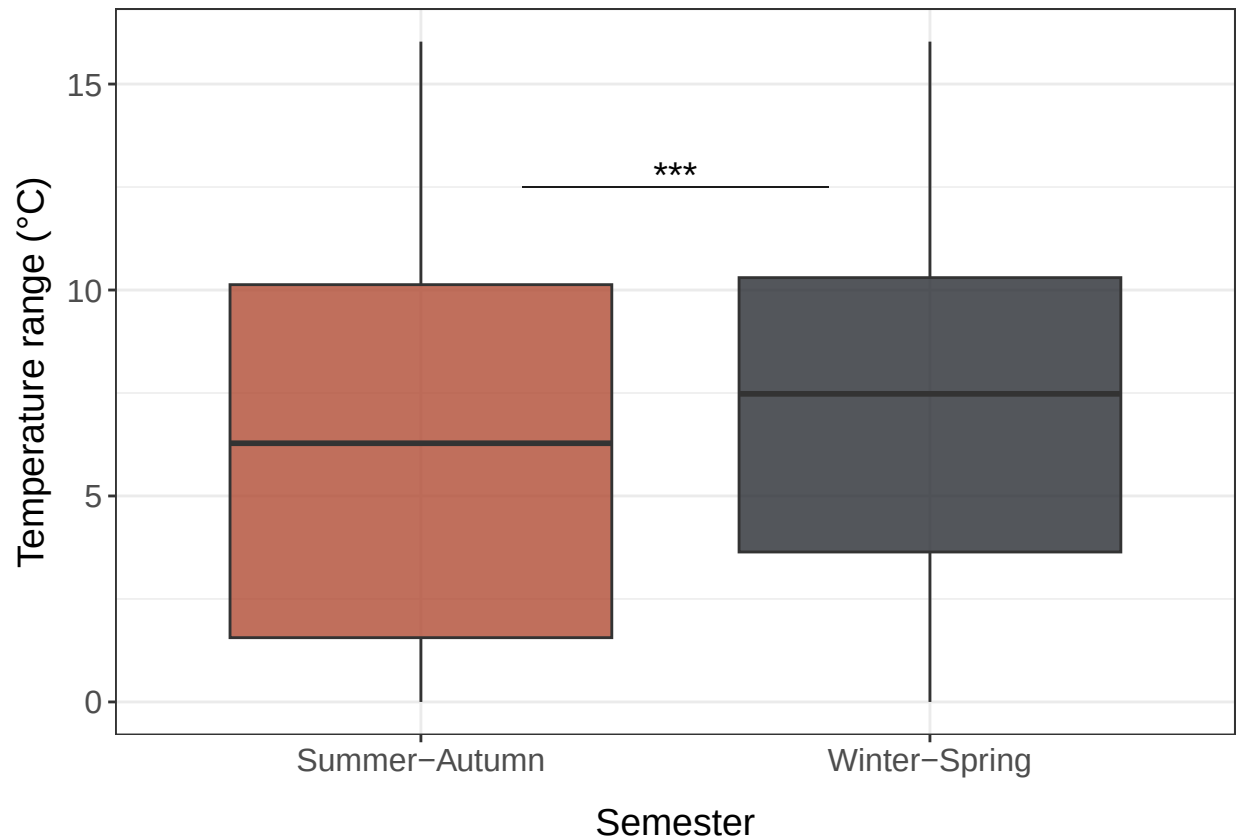
5. Create boxplot comparing Semester's temperature vs ASVs optimal temperature

```
text_size <- 12

range_temp_boxplot <- ggplot(asvs2range, aes(x = Community,
                                             y = asvs_temp_range, fill = Community)) +
  # add a line at y = 1
  geom_boxplot(alpha = 0.8) +
  theme_minimal() +
  xlab("Semester") +
  ylab("Temperature range (°C)") +
  scale_fill_manual(values = c("#b14b34", "#292c33"),
                    labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring"),
                    name = "Semester") +
  scale_x_discrete(labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring")) +
  theme_bw() +
  theme(
    axis.text.x = element_text(size = text_size),
    axis.text.y = element_text(size = text_size),
    axis.title.x = element_text(size = text_size + 2, margin = unit(c(4,0,0,0), "mm")),
    axis.title.y = element_text(size = text_size + 2, margin = unit(c(0,2,0,0), "mm")),
    strip.text = element_text(size = text_size + 4),
    strip.background = element_blank(),
    # increase legend text
    legend.text = element_text(size = text_size),
    legend.title = element_text(size = text_size + 4)
  ) +
  # remove legend
  theme(legend.position = "none")

range_temp_boxplot_wtext <- range_temp_boxplot +
  # add text to the plot
  geom_segment(aes(x = 1.2, y = 12.5, xend = 1.8, yend = 12.5),
              color = "gray10", alpha = 0.7, linewidth = 0.1) +
  annotate("text", x = 1.5, y = 12.8, label = "***", size = 5)

range_temp_boxplot_wtext
```



6. Run Welch's ANOVA comparing the asvs_temp_range

```
wtest_range_temp <- oneway.test(asvs_temp_range ~ Community,
                                data = asvs2range, var.equal = FALSE)
```

```
wtest_range_temp
```

```
##
## One-way analysis of means (not assuming equal variances)
##
## data: asvs_temp_range and Community
## F = 144.2, num df = 1, denom df = 11595, p-value < 2.2e-16
```

9. Print session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
```

```

## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C                LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
## [6] LC_MESSAGES=en_US.UTF-8    LC_PAPER=en_US.UTF-8       LC_NAME=C                  LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] fANCOVA_0.6-1      maps_3.4.2          corrgram_1.14       ggpubr_0.6.0        ggcorrplot_0.1.4.1
## [8] permute_0.9-7      lubridate_1.9.3     forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [15] tidyr_1.3.1        tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.5      xfun_0.48           rstatix_0.7.2       tzdb_0.4.0          vctrs_0.6.5         tools
## [9] fansi_1.0.6       highr_0.11          cluster_2.1.6       pkgconfig_2.0.3     Matrix_1.7-0        lifecycle
## [17] tinytex_0.53      munsell_0.5.1       carData_3.0-5       htmltools_0.5.8.1   yaml_2.3.10         Formula
## [25] car_3.1-3         MASS_7.3-61         abind_1.4-8         nlme_3.1-166        tidyselect_1.2.1    digest
## [33] splines_4.4.2     cowplot_1.1.3       fastmap_1.2.0       colorspace_2.1-1    cli_3.6.3           magrittr
## [41] withr_3.0.1       scales_1.3.0        backports_1.5.0     bit64_4.5.2         timechange_0.3.0    rmarkdown
## [49] hms_1.1.3         evaluate_1.0.1      knitr_1.48          mgcv_1.9-1          rlang_1.1.4         glue
## [57] R6_2.5.1

```